

1 Inline and display mathematics

The formula is now displayed separately:

$$y = mx + c$$

The same formula in display mode:

$$y = mx + c$$

Superscripts and subscripts: a^b , a_b , x^2 , x_i .

Greek letters and mathematical functions: α , β , θ , Γ , Δ , $y = 2 \sin \theta^2$, $\log x$.

2 Numbered equation

An integral with an automatically generated number:

$$(1) \quad \int_{-\infty}^{+\infty} e^{-x^2} dx$$

3 Aligned equations

A recurrence without equation numbers:

$$Q_{n,0} = 1 \quad Q_{0,k} = [k = 0];$$

$$Q_{n,k} = Q_{n-1,k} + Q_{n-1,k-1} + \binom{n}{k}, \quad \text{for } n, k > 0.$$

Several columns of aligned equations:

$$\begin{array}{lll} a = b + 1 & c = d + 2 & e = f + 3 \\ r = s^2 & t = u^3 & v = w^4 \end{array}$$

4 Matrices

Three matrix styles:

$$\begin{array}{ccc} a & b & c \\ d & e & f \end{array} \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \quad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

Right-aligned matrix columns using mathtools:

$$\begin{pmatrix} 10 & 11 \\ 1 & 2 \\ -5 & -6 \end{pmatrix}$$

5 Other multiline environments

Centered equations without alignment:

- (2) $P(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f$
(3) $x^2 + x = 10$

A single expression split across several lines:

$$\begin{aligned} &(a + b + c + d)x^5 + (b + c + d + e)x^4 \\ &+ (c + d + e + f)x^3 + (d + e + f + a)x^2 + (e + f + a + b)x \\ &+ (f + a + b + c) \end{aligned}$$

An aligned block inside a larger expression:

$$\begin{cases} a = b \\ c = d \end{cases}$$

6 Mathematical fonts

ABC *ABC* **ABC** ABC **ABC** **ABC**

Comparing several ways of writing a word:

size $\neq$ *size* $\neq$ size

Nested font commands:

ABC **ABC**

7 Bold mathematics

Normal weight: $(x + y)(x - y) = x^2 - y^2$.

A bold expression: $(\mathbf{x} + \mathbf{y})(\mathbf{x} - \mathbf{y}) = \mathbf{x}^2 - \mathbf{y}^2$.

Individual bold Latin letters: $(x + \mathbf{y})(x - \mathbf{y}) = x^2 - \mathbf{y}^2$.

Bold Greek letters using bm: $\alpha + \boldsymbol{\alpha} < \beta + \boldsymbol{\beta}$.